A Small Test of Cache-Aware RISC-V Scheduling

Synthetic academic-paper fixture for verifying the MVP 2 parser and extractor. Not a published paper.

Abstract

This synthetic paper studies a lightweight scheduling policy for a RISC-V teaching simulator. Existing classroom examples often ignore cache misses and therefore provide an incomplete explanation of execution-time variation. This work proposes a simple cache-aware scheduling heuristic and evaluates it on a small set of generated workloads. Results show a modest reduction in simulated stalls compared with a round-robin baseline.

1 Introduction

Computer architecture students need practical ways to understand how scheduling decisions interact with memory behavior. A key challenge is that simple scheduling examples do not represent cache effects, which can make the observed performance difficult to explain. This paper presents a cache-aware scheduling heuristic for a teaching simulator. We investigate whether basic cache-miss information can improve simulated instruction throughput without adding substantial policy complexity.

2 Methodology

The proposed method assigns each runnable task a score based on its recent cache-miss rate and estimated remaining work. The scheduler selects the task with the lowest combined score at each scheduling interval. We implement the policy in a small RISC-V teaching simulator and compare it with round-robin scheduling. The experiments use a generated dataset of 24 deterministic workloads, including sequential, random, and mixed memory-access patterns.

3 Results and Evaluation

The evaluation shows that the cache-aware policy reduces simulated memory stalls by 8 percent on the mixed workload group. The results also show that the policy does not improve every workload: sequential workloads have nearly identical stall counts under both policies. These findings suggest that cache-miss information can help in some teaching scenarios, but the benefit depends on the workload.

4 Conclusion

This paper introduced a cache-aware scheduling heuristic for a RISC-V teaching simulator. In the evaluated synthetic workloads, the heuristic reduced simulated stalls for mixed access patterns while adding a small amount of scheduling complexity. The experiment demonstrates a useful teaching example rather than evidence of a production-ready scheduler.

5 Limitations

The evaluation uses generated workloads and a simplified cache model, so the results may not generalize to physical RISC-V hardware. The experiment also uses a fixed number of tasks and does not measure energy consumption.

6 Future Work

Future work should evaluate the policy on real hardware and larger public workload suites. Further investigation is needed to understand how the heuristic behaves when task priorities change dynamically.